

Andrés Felipe Blanco Sierra

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EDUCATION

Universidad EAN

B.S. Systems Engineering – 7th semester

Bogotá, Colombia

2023 – 2027 (expected)

LEADERSHIP & HONORS

Student Representative – Faculty of Engineering

Student voice in front of the faculty; liaises between students and faculty leadership

- Represent the Faculty of Engineering, channeling academic concerns and student initiatives

Universidad EAN

2026

Workshop Instructor and Jury – The Code Game

National programming competition for school students (Scratch, Python, Java)

- Delivered workshops and judged student-built video games in the programming competition

Colegio Tilatá

2026

University Choir – ASCUN Culture Choir Festival

Choral competition of the Colombian Association of Universities (ASCUN)

- Awarded 3rd place in the university choral competition

Universidad EAN

2025

PROJECTS

WHAT? – Learn Languages with Songs | *React 19, TypeScript, FastAPI, Whisper, TanStack Query, Tailwind 4*

2026

- Built a full-stack app that downloads songs (yt-dlp), transcribes them with Whisper large-v3 and syncs them with Genius lyrics for language learning
- Developed an async backend (SQLAlchemy + SQLite) with a React 19, TanStack Query and Tailwind 4 frontend, plus Anki export

Portrait Dataset Builder | *Python, Typer, OpenCV, InsightFace, CLIP, Hydra*

2026

- Built a CLI that ingests images/video, detects and crops faces with InsightFace and deduplicates with perceptual hashing
- Implemented semantic classification with CLIP and vector search (usearch); Hydra-driven pipeline, async tests and strict linting (ruff/black/mypy)

OpenCode Telegram Controller | *Python, aiogram, SQLite, systemd*

2026

- Built a Telegram bot that runs AI (opencode) tasks with a task queue, controlled concurrency and graceful cancellation
- Implemented allowlist-based security, deterministic summaries and 110 automated tests

Guitar Hero Controller | *C++, Arduino, uinput, termios*

2026

- Designed a physical controller: Arduino firmware (5 frets + joystick, 4-byte serial protocol) and a C++ driver emulating a keyboard/joystick on Linux via uinput

CP2077 UI – Cyberpunk 2077 Replica | *React 19, TypeScript, Vite, Tailwind*

2026

- Replicated the game interface (Hub, character, cyberware, inventory and stats) with a custom router and virtualized lists

Physical Evolution System (GYM OS) | *Python, SQLite, inotify, pytest*

2026

- Developed a backend that parses an Obsidian vault into SQLite and computes progression, PRs and gamification (XP/levels) with an automatic inotify watcher
- 88 automated tests; feeds live desktop wallpapers in real time

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Bash, C++

Backend: FastAPI, SQLAlchemy, REST APIs, Node.js/Express, Laravel

Frontend: React 19, TypeScript, Vite, Tailwind CSS, TanStack Query, Next.js

AI / ML: Whisper, CLIP, OpenCV, InsightFace, Tesseract OCR, embeddings, scikit-learn

Systems: Linux (Arch/Hyprland), systemd, Docker, Git, uinput, Arduino

ADDITIONAL

Languages: Spanish (native), English (B2+ – professional working proficiency; TOEFL in preparation)

Honors: Artistic excellence scholarship – EAN Philharmonic Orchestra; Best student in cultural and wellness classes, EAN (2024-1 to 2025-2); 3rd place ASCUN Culture Choir Festival 2025